Polar Form, De Moivre's Theorem & Roots of Unity

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Use De Moivre's theorem to prove that $\cos 3\theta = 4\cos^3\theta - 3\cos\theta$. Hence solve $4x^3 - 3x = 0$ on $x \in [-1, 1]$.

1. Expand $(\cos\theta + i\sin\theta)^3$ via Binomial Theorem: $(\cos\theta + i\sin\theta)^3 = \cos^3\theta + 3i\cos^2\theta\sin\theta - 3\cos\theta\sin^2\theta - i\sin^3\theta$. 2. By De Moivre's Theorem, this equals $\cos 3\theta + i\sin 3\theta$. Equate real parts: $\cos 3\theta = \cos^3\theta - 3\cos\theta\sin^2\theta$. Substitute $\sin^2\theta = 1 - \cos^2\theta$.

2 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Consider the function $f(x) = x^2 - 5x + 6$. The graph of this function is a parabola. Determine the x-coordinates of the points where the graph intersects the x-axis.

To find the x-coordinates where the graph intersects the x-axis, we need to find the roots of the function, i.e., where $f(x) = 0$. We set the function equal to zero: $x^2 - 5x + 6 = 0$. This is a quadratic equation that can be factored. We are looking for two numbers that multiply to 6 and add to -5. These numbers are -2 and -3. So, the equation can be factored as: $(x - 2)(x - 3) = 0$. For the product of two factors to be zero, at least one of the factors must be zero. Therefore, $x - 2 = 0$ or $x - 3 = 0$. Solvi.

3 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Consider the function $f(x) = ax^2 + bx + c$. The graph of $f(x)$ is a parabola that passes through the points $(-1, 12)$, $(1, 2)$, and $(2, 9)$. Find the value of the parameter b.

Let the function be $f(x) = ax^2 + bx + c$. We are given three points that lie on the graph of $f(x)$: $(-1, 12)$, $(1, 2)$, and $(2, 9)$. Substituting these points into the function, we get a system of three linear equations with three unknowns a, b, and c: 1. For $(-1, 12)$: $a(-1)^2 + b(-1) + c = 12 \Rightarrow a - b + c = 12$ 2. For $(1, 2)$: $a(1)^2 + b(1) + c = 2 \Rightarrow a + b + c = 2$ 3. For $(2, 9)$: $a(2)^2 + b(2) + c = 9 \Rightarrow 4a + 2b + c = 9$ We can solve this system of equations. Subtract.

4 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Two straight paths in 3D space are modeled by lines $L_1 : \mathbf{r}_1 = (1; 2; 0) + \lambda(1; -1; 2)$ and $L_2 : \mathbf{r}_2 = (0; 1; -1) + \mu(2; 1; 1)$. Show that L_1 and L_2 are skew, and find the shortest perpendicular distance between them.

1. Non-parallel test: Direction vectors $\mathbf{d}_1 = \langle 1, -1, 2\rangle$ and $\mathbf{d}_2 = \langle 2, 1, 1\rangle$ are not scalar multiples ($1/2 eq -1/1$). 2. Normal direction perpendicular to both lines:

$\mathbf{n} = \mathbf{d}_1 \times \mathbf{d}_2 = |\mathbf{i}, \mathbf{j}, \mathbf{k}; 1, -1, 2; 2, 1, 1| = \mathbf{i}(-1 - 2) - \mathbf{j}(1 - 4) + \mathbf{k}(1 - (-2)) = -3\mathbf{i} + 3\mathbf{j} + 3\mathbf{k} \propto (-1; 1; 1)$. Magnitude.

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5 [Foundational] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

(IB AI HL) A market transition matrix $T = (0.7, 0.2; 0.3, 0.8)$ represents weekly customer movement between Brand A and Brand B. Find: (a) the probability that a customer initially with Brand A is with Brand B after 2 weeks; (b) the long-term steady-state market distribution.

(a) State distribution after 2 weeks is $\mathbf{p}_2 = T^2\mathbf{p}_0$. With $\mathbf{p}_0 = (1; 0)$: $T^2 = (0.7, 0.2; 0.3, 0.8)(0.7, 0.2; 0.3, 0.8) = (0.49 + 0.06, 0.14 + 0.16; 0.21 + 0.24, 0.06 + 0.64) = (0.55, 0.30; 0.45, 0.70)$. Thus $\mathbf{p}_2 = (0.55; 0.45)$, so the probability of being with Brand B after 2 weeks is 0.45 (45%). (b) Steady-state vector $\mathbf{s} = (s_1; s_2)$ satisfies $(T - I)\mathbf{s} = \mathbf{0}$.

6 [Foundational] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Determine the equation of the oblique asymptote to the graph of $f(x) = \frac{2x^2 - 3x + 4}{x - 1}$.

To find the equation of the oblique asymptote, divide the numerator by the denominator using polynomial long division or synthetic division: $2x^2 - 3x + 4, = 2x(x - 1) - x + 4; , = 2x(x - 1) - 1(x - 1) + 3; , = (2x - 1)(x - 1) + 3$ Dividing through by $x - 1$ yields: $f(x) = 2x - 1 + \frac{3}{x-1}$ As $x \to \pm\infty$, the remainder term $\frac{3}{x-1} \to 0$. Therefore, the linear quotient gives the oblique asymptote: $y = 2x - 1$ - $y = 2x - 3$ arises from mistakenly dividing each of the f.

7 [Foundational] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Use trigonometric substitution to evaluate the indefinite integral $\int \frac{1}{(4 + x^2)^{3/2}}\,dx$.

1. Substitute $x = 2\tan\theta \Rightarrow dx = 2\sec^2\theta\,d\theta$. Then $4 + x^2 = 4 + 4\tan^2\theta = 4\sec^2\theta$, so: $(4 + x^2)^{3/2} = (4\sec^2\theta)^{3/2} = 8\sec^3\theta$. 2. Substitute into the integral: $\int \frac{2\sec^2\theta}{8\sec^3\theta}\,d\theta = \frac{1}{4}\int \frac{1}{\sec\theta}\,d\theta = \frac{1}{4}\int \cos\theta\,d\theta = \frac{1}{4}\sin\theta + C$. 3. Express $\sin\theta$ in terms of x: From $\tan\theta = \frac{x}{2}$, construct a right triangle with opposite side x, adjacen.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

(IB AI HL) A communications network with 6 nodes $\{A, B, C, D, E, F\}$ has weighted edges: AB=4, AC=2, BC=1, BD=5, CD=8, CE=10, DE=2, DF=6, EF=3. Use Kruskal's algorithm to determine the minimum spanning tree, stating the order of edges added, and find the total tree weight.

1. Sort all edges in non-decreasing order of weight: BC (1), AC (2), DE (2), EF (3), AB (4), BD (5), DF (6), CD (8), CE (10). 2. Apply Kruskal's algorithm (selecting $V - 1 = 6 - 1 = 5$ edges without cycles): - Add BC (weight 1) [Tree components: {B, C}] - Add AC (weight 2) [Merges A: {A, B, C}] - Add DE (weight 2) [Component: {D, E}] - Add EF (weight 3) [Merges F: {D, E, F}] - Consider AB (weight 4): A and B are already connected in {A, B, C} (would create cycle AB-BC-CA), so SKIP AB. - Add BD (weight 5): Connects.

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9 **[Intermediate]** • **Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)**

Consider the function $f(x) = \frac{\sin(2x)}{\cos(x)}$. For the domain $x \in [0, 2\pi]$ such that $\cos(x) \neq 0$, find the exact value of the definite integral $\int_0^{\pi/2} f(x)dx$.

First, we simplify the function $f(x)$. Using the double angle identity for sine, $\sin(2x) = 2\sin(x)\cos(x)$. Therefore, $f(x) = \frac{2\sin(x)\cos(x)}{\cos(x)}$. For $\cos(x) \neq 0$, we can cancel the $\cos(x)$ terms, so $f(x) = 2\sin(x)$. Now we need to evaluate the definite integral: $\int_0^{\pi/2} 2\sin(x)dx$. The antiderivative of $2\sin(x)$ is $-2\cos(x)$. Evaluating the definite integral using the Fundamental Theorem of Calculus: $[-2\cos(x)]_0^{\pi/2} = -2\cos(\pi/2) - (-2\cos(0))$.

10 **[Intermediate]** • **Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)**

Using the binomial series for $(1 - x^2)^{-1/2}$, derive the Maclaurin series expansion for $f(x) = \arcsin x$ up to and including the x^5 term. State its radius of convergence.

1. Recall the general binomial expansion $(1 + u)^p = 1 + pu + \frac{p(p-1)}{2!}u^2 + \ldots$. Here $u = -x^2$ and $p = -1/2$: $(1 - x^2)^{-1/2} = 1 + (-\frac{1}{2})(-x^2) + \frac{(-1/2)(-3/2)}{2}(-x^2)^2 + O(x^6) = 1 + \frac{1}{2}x^2 + \frac{3}{8}x^4 + O(x^6)$. 2. Since $\arcsin x = \int_0^x \frac{1}{\sqrt{1-t^2}}dt$, integrate term-by-term: $\arcsin x = \int_0^x (1 + \frac{1}{2}t^2 + \frac{3}{8}t^4 + \ldots)dt = x + \frac{1}{6}x^3 + \frac{3}{40}x^5 + \ldots$ 3. The binomial series converge.

11 **[Intermediate]** • **Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)**

A survey was conducted on the number of hours students spent studying per week. The results are summarized in the histogram below. The total number of students surveyed is 120. Let x represent the number of students who studied for more than 3 hours but at most 4 hours. Find the value of x.

The histogram shows the frequency distribution of study hours. The total number of students surveyed is 120. The number of students in each interval is represented by the height of the bars multiplied by the width of the interval. Interval (0, 1]: Frequency = 15 Interval (1, 2]: Frequency = 25 Interval (2, 3]: Frequency = 30 Interval (3, 4]: Frequency = x Interval (4, 5]: Frequency = 20 The sum of the frequencies must equal the total number of students: $15 + 25 + 30 + x + 20 = 120\ 90 + x = 120\ x = 120 - 90$.

12 **[Intermediate]** • **Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)**

Decompose the rational expression $\frac{5x^2 - 3x + 2}{(x-1)(x^2+1)}$ into partial fractions.

1. Set up the partial fraction template with irreducible quadratic: $\frac{5x^2 - 3x + 2}{(x-1)(x^2+1)} = \frac{A}{x-1} + \frac{Bx + C}{x^2+1}$. 2. Multiply through by $(x - 1)(x^2 + 1)$: $5x^2 - 3x + 2 = A(x^2 + 1) + (Bx + C)(x - 1)$. 3. Substitute $x = 1$ (cover-up): $5(1)^2 - 3(1) + 2 = A(1 + 1) + 0 \Rightarrow 4 = 2A \Rightarrow A = 2$. 4. Compare coefficients: Coefficient of x^2: $A + B = 5 \Rightarrow 2 + B = 5 \Rightarrow B = 3$. Constant term: $A - C = 2 \Rightarrow 2 - C = 2 \Rightarrow C = 0$. D.

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13 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

The derivative of the function $f(x) = 3x^2 - 5x + 2$ is given by $f'(x)$. Find the value of $f'(2)$.

To find the derivative of the function $f(x) = 3x^2 - 5x + 2$, we use the power rule for differentiation, which states that $\frac{d}{dx}(ax^n) = nax^{n-1}$. Applying this rule to each term: 1. The derivative of $3x^2$ is $2 \cdot 3x^{2-1} = 6x$. 2. The derivative of $-5x$ is $1 \cdot (-5)x^{1-1} = -5x^0 = -5$. 3. The derivative of a constant, $+2$, is 0. So, the derivative of $f(x)$ is $f'(x) = 6x - 5$. Now, we need to find the value of $f'(2)$. We substitute $x = 2$ into the expression for $f'(x)$.

14 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

A random variable X follows a binomial distribution $B(n, p)$. The number of trials is $n = 10$ and the probability of success in a single trial is p. If the mean of X is 4, find the variance of X.

For a binomial distribution $X \sim B(n, p)$, the mean is given by $E(X) = np$ and the variance is given by $Var(X) = np(1 - p)$. We are given that $n = 10$ and $E(X) = 4$. Using the formula for the mean, we have $10p = 4$. Solving for p, we get $p = \frac{4}{10} = 0.4$. Now we can find the variance using the formula $Var(X) = np(1 - p)$. Substituting the values of n and p, we get $Var(X) = 10 \times 0.4 \times (1 - 0.4)$. $Var(X) = 10 \times 0.4 \times 0.6$. $Var(X) = 4 \times 0.6$. $Var(X) = 2.4$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 6.03 Markov Chains (AI HL)

Find all complex solutions to $z^4 + 16 = 0$ in exact rectangular form $a + bi$ using De Moivre's roots theorem.

Step 1: Express -16 in polar modulus-argument form: $z^4 = -16 = 16(\cos\pi + i\sin\pi)$. Here $r = 16$, $\theta = \pi$, and $n = 4$. Step 2: Apply n-th roots formula: $w_k = \sqrt[4]{16}\left[\cos\left(\frac{\pi+2\pi k}{4}\right) + i\sin\left(\frac{\pi+2\pi k}{4}\right)\right] = 2\left[\cos\left(\frac{(2k+1)\pi}{4}\right) + i\sin\left(\frac{(2k+1)\pi}{4}\right)\right],\ k \in \{0, 1, 2, 3\}$. Step 3: Evaluate for each k:$k = 0$.

16 [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 1.01 Proof by Mathematical Induction

Find all distinct points of intersection between the two polar curves $r = 2\sin\theta$ and $r = 2\cos\theta$ on $0 \le \theta < 2\pi$.

Equate the two equations: $2\sin\theta = 2\cos\theta \Rightarrow \tan\theta = 1$. On $[0, 2\pi)$, $\theta = \frac{\pi}{4}$ or $\theta = \frac{5\pi}{4}$.For $\theta = \frac{\pi}{4}$: $r = 2\sin\left(\frac{\pi}{4}\right) = \sqrt{2}$, giving point $\left(\sqrt{2}, \frac{\pi}{4}\right)$.For $\theta = \frac{5\pi}{4}$: $r = 2\sin\left(\frac{5\pi}{4}\right) = -\sqrt{2}$. However, $\left(-\sqrt{2}, \frac{5\pi}{4}\right)$ represents the identical geometric point as $\left(\sqrt{2}, \frac{\pi}{4}\right)$.Next, check the pole.

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17 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 1.02 Complex Numbers: Cartesian Form & Algebra

[Polar Coordinates, Complex Numbers & De Moivre's Theorem · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

Substitute $t = 8$: $g(8) = 5(8) + 5 = \mathbf{45}$. Setting $5t + 5 = 50$ gives $5t = 45 \Rightarrow t = \mathbf{9}$.

18 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 1.04 Partial Fractions

Use the substitution $x = 2\sin\theta$ to evaluate the exact definite integral $\int_0^1 \sqrt{4 - x^2}\, dx$.

1. Substitute $x = 2\sin\theta \Rightarrow dx = 2\cos\theta\, d\theta$. When $x = 0$, $\theta = 0$. When $x = 1$, $2\sin\theta = 1 \Rightarrow \sin\theta = \frac{1}{2} \Rightarrow \theta = \frac{\pi}{6}$. 2. Rewrite integrand: $\sqrt{4 - x^2} = \sqrt{4 - 4\sin^2\theta} = 2\sqrt{\cos^2\theta} = 2\cos\theta$ (since $\cos\theta > 0$ on $[0, \pi/6]$). 3. Substitute into the integral.

19 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 1.05 Systems of Linear Equations (AA HL)

Find the exact range of $f(x) = \frac{x}{x^2 + 1}$ for $x \in \mathbb{R}$.

Let $y = \frac{x}{x^2 + 1}$. Multiply both sides by $x^2 + 1$ (valid since $x^2 + 1 > 0$ for all $x \in \mathbb{R}$): $y(x^2 + 1) = x \Rightarrow yx^2 - x + y = 0$. If $y = 0$, $-x = 0 \Rightarrow x = 0$, so 0 is in the range. For $yeq0$, this is a quadratic equation in x. For real x to exist, the discriminant Δ must be non-negative: $\Delta = (-1)^2 - 4(y)(y) = 1 - 4y^2 \geq 0$. Solve the inequality: $4y^2 \leq 1 \Rightarrow y^2 \leq \frac{1}{4} \Rightarrow -\frac{1}{2} \leq y \leq \frac{1}{2}$. Thu.

20 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 2.01 Modulus Functions & the Factor/Remainder Theorem

A weighted network has edges: DE = 1, BC = 2, AB = 3, CD = 4, BD = 5, CE = 6, AC = 7, AD = 8. Use Kruskal's algorithm to determine the minimum spanning tree and compute its total weight.

1. Sort all edges in non-decreasing order of weight: DE (1), BC (2), AB (3), CD (4), BD (5), CE (6), AC (7), AD (8). 2. Select edges avoiding cycles: - Add DE (weight 1) [Vertices connected: {D, E}] - Add BC (weight 2) [Vertices connected: {D, E}, {B, C}] - Add AB (weight 3) [Vertices connected: {D, E}, {A, B, C}] - Add CD (weight 4) [Merges components: {A, B, C, D, E} - all 5 vertices connected!] 3. Check edge count: $V - 1 = 5 - 1 = 4$ edges selected. 4. Total MST Weight: $1 + 2 + 3 + 4 = 10$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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21 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Express $z = -2 - 2\sqrt{3}i$ in Euler's exponential form $re^{i\theta}$ with $\theta \in (-\pi, \pi]$.

a. $4e^{-i2\pi/3}$ ✓

b. $4e^{i4\pi/3}$

c. $2e^{-i2\pi/3}$

d. $4e^{-i\pi/3}$

$r = \sqrt{(-2)^2 + (-2\sqrt{3})^2} = \sqrt{4 + 12} = 4$. In Quadrant III ($x < 0, y < 0$), $\theta = -\pi + \arctan(\sqrt{3}) = -\pi + \frac{\pi}{3} = -\frac{2\pi}{3}$. Thus $z = 4e^{-i2\pi/3}$.

22 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Convert $z = 6e^{i3\pi/4}$ into exact Cartesian form $x + iy$.

a. $-3\sqrt{2} + 3\sqrt{2}i$ ✓

b. $3\sqrt{2} - 3\sqrt{2}i$

c. $-3 + 3\sqrt{3}i$

d. $-3\sqrt{2} - 3\sqrt{2}i$

$x = 6\cos(3\pi/4) = 6(-\sqrt{2}/2) = -3\sqrt{2}$. $y = 6\sin(3\pi/4) = 6(\sqrt{2}/2) = 3\sqrt{2}$.

23 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

If $z_1 = 3e^{i\pi/3}$ and $z_2 = 2e^{i\pi/6}$, evaluate the quotient $\frac{z_1}{z_2}$ in exponential form.

a. $\frac{3}{2}e^{i\pi/6}$ ✓

b. $\frac{3}{2}e^{i\pi/2}$

c. $6e^{i\pi/6}$

d. $1.5e^{-i\pi/6}$

$\frac{z_1}{z_2} = \frac{r_1}{r_2}e^{i(\theta_1 - \theta_2)} = \frac{3}{2}e^{i(\pi/3 - \pi/6)} = \frac{3}{2}e^{i\pi/6}$.

24 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Use De Moivre's Theorem to evaluate $(1 + i\sqrt{3})^6$.

a. 64 ✓

b. -64

c. $64i$

d. $-64i$

In polar form, $1 + i\sqrt{3} = 2e^{i\pi/3}$. By De Moivre's Theorem: $(2e^{i\pi/3})^6 = 2^6 e^{i6(\pi/3)} = 64e^{i2\pi} = 64(\cos 2\pi + i\sin 2\pi) = 64(1 + 0) = 64$.

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25 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Express $\cos(3\theta)$ in terms of $\cos\theta$ using De Moivre's Theorem $(\cos\theta + i\sin\theta)^3$.

a. $4\cos^3\theta - 3\cos\theta$ ✓

b. $3\cos^3\theta - 4\cos\theta$

c. $\cos^3\theta - 3\cos\theta$

d. $4\cos^3\theta + 3\cos\theta$

Expand $(\cos\theta + i\sin\theta)^3 = \cos^3\theta + 3i\cos^2\theta\sin\theta - 3\cos\theta\sin^2\theta - i\sin^3\theta$. Equating real parts: $\cos(3\theta) = \cos^3\theta - 3\cos\theta(1 - \cos^2\theta) = 4\cos^3\theta - 3\cos\theta$.

26 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Find all cube roots of $8i$ in polar form.

a. $2e^{i\pi/6}, 2e^{i5\pi/6}, 2e^{-i\pi/2}$ ✓

b. $2e^{i\pi/3}, 2e^{i\pi}, 2e^{i5\pi/3}$

c. $8e^{i\pi/6}, 8e^{i5\pi/6}, 8e^{-i\pi/2}$

d. $2,\ -1 + i\sqrt{3},\ -1 - i\sqrt{3}$

$8i = 8e^{i\pi/2}$. The cube roots have magnitude $8^{1/3} = 2$ and arguments $\frac{\pi/2 + 2k\pi}{3} = \frac{\pi}{6} + \frac{2k\pi}{3}$ for $k = 0, 1, 2$, giving $\frac{\pi}{6}, \frac{5\pi}{6}, \frac{9\pi}{6} \equiv -\frac{\pi}{2}$.

27 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

How are the n distinct nth roots of any complex number $z = re^{i\theta}$ distributed on the Argand plane?

a. **They lie on a circle of radius $r^{1/n}$ centered at the origin, evenly spaced by $\frac{2\pi}{n}$ radians** ✓

b. They lie along a straight line through the origin

c. They form an ellipse with eccentricity $1/n$

d. They cluster exclusively in Quadrant I

All nth roots share the identical modulus $r^{1/n}$ and their arguments differ by constant increments of $\frac{2\pi}{n}$, forming a regular n-sided polygon centered at the origin.

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28 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

What is the sum of all n distinct nth roots of any complex number z ($n \geq 2$)?

a. 0 ✓

b. 1

c. n

d. z

By Vieta's formulas applied to $w^n - z = 0$, the coefficient of w^{n-1} is zero, so the sum of all roots is strictly zero.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

If $\omega = e^{i2\pi/3}$ is a primitive complex cube root of unity, what is the value of $1 + \omega + \omega^2$?

a. 0 ✓

b. 1

c. -1

d. 3

For any nth root of unity $\omega \neq 1$, $\sum_{k=0}^{n-1} \omega^k = \frac{\omega^n - 1}{\omega - 1} = \frac{1-1}{\omega-1} = 0$. For $n = 3$, $1 + \omega + \omega^2 = 0$.

30 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Factor $z^4 + 1$ into two real quadratic factors.

a. $(z^2 - \sqrt{2}z + 1)(z^2 + \sqrt{2}z + 1)$ ✓

b. $(z^2 + 1)^2$

c. $(z^2 - 1)(z^2 + 1)$

d. $(z^2 - 2z + 1)(z^2 + 2z + 1)$

Complete the square: $z^4 + 1 = (z^2+1)^2 - 2z^2 = (z^2+1)^2 - (\sqrt{2}z)^2 = (z^2 - \sqrt{2}z + 1)(z^2 + \sqrt{2}z + 1)$.

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31 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

The four 4th roots of unity ($z^4 = 1$) form the vertices of which geometric shape on the Argand plane?

a. A square inscribed in the unit circle with vertices at $\pm 1, \pm i$ ✓

b. An equilateral triangle

c. A regular hexagon

d. A rectangle of perimeter 4

The solutions to $z^4 = 1$ are $1, i, -1, -i$. Plotted on the Argand diagram, they form a square of side length $\sqrt{2}$ inscribed in the unit circle.

32 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

For $x^2 - 6x + 5 = 0$, find the sum of the roots.

a. 6 ✓

b. 11

c. 5

d. 3

Working through this complex-numbers step: $\text{Sum of roots} = -\frac{b}{a} = -(-6) = 6$

33 [Intermediate] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Let the complex number z be represented in polar form as $z = r(\cos\theta + i\sin\theta)$. Suppose z satisfies the equation $z^3 = -64i$. If z is restricted to the fourth quadrant (i.e., $\frac{3\pi}{2} < \arg(z) < 2\pi$), find the real part of z.

a. 4

b. -4

c. $2\sqrt{3}$ ✓

d. $-2\sqrt{3}$

We are given the equation $z^3 = -64i$. First, express $-64i$ in polar form. The modulus is $|-64i| = 64$. The argument is $\arg(-64i) = \frac{3\pi}{2}$ (or $-\frac{\pi}{2}$). So, $-64i = 64\left(\cos\frac{3\pi}{2} + i\sin\frac{3\pi}{2}\right)$. Let $z = r(\cos\phi + i\sin\phi)$. Then $z^3 = r^3(\cos(3\phi) + i\sin(3\phi))$. Equating the polar forms, we have $r^3 = 64$ and $3\phi = \frac{3\pi}{2} + 2k\pi$ for some integer k. From $r^3 = 64$, we get $r = 4$ (since r is a non-negative real number). Fro.

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34 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Let the complex number z_1 be represented in polar form as $z_1 = r_1(\cos\theta_1 + i\sin\theta_1)$, where $r_1 = 4$ and $\theta_1 = \frac{\pi}{3}$. Let another complex number, z_2, be represented as $z_2 = 2(\cos\theta_2 + i\sin\theta_2)$. If $z_1 \cdot z_2 = 8(\cos\alpha + i\sin\alpha)$ and $\alpha = \frac{5\pi}{6}$, find the value of θ_2 in radians, given that $0 \le \theta_2 < 2\pi$.

a. $\frac{\pi}{2}$ ✓

b. $\frac{2\pi}{3}$

c. $\frac{5\pi}{6}$

d. $\frac{\pi}{6}$

We are given two complex numbers in polar form: $z_1 = r_1(\cos\theta_1 + i\sin\theta_1)$ and $z_2 = r_2(\cos\theta_2 + i\sin\theta_2)$. We are provided with $r_1 = 4$ and $\theta_1 = \frac{\pi}{3}$. For z_2, we are given $r_2 = 2$. The product of two complex numbers in polar form is given by $z_1 \cdot z_2 = r_1 r_2(\cos(\theta_1 + \theta_2) + i\sin(\theta_1 + \theta_2))$. We are given that $z_1 \cdot z_2 = 8(\cos\alpha + i\sin\alpha)$, where $\alpha = \frac{5\pi}{6}$. Equating the magnitudes, we.

35 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

Consider the complex number $z = \left(\frac{\sqrt{3}}{2} + \frac{1}{2}i\right)^{10}$. If z is expressed in polar form as $r(\cos\theta + i\sin\theta)$, where $r > 0$ and $0 \le \theta < 2\pi$, find the exact values of r and θ.

a. $r = 1, \theta = \frac{5\pi}{3}$ ✓

b. $r = 1, \theta = \frac{4\pi}{3}$

c. $r = 10, \theta = \frac{5\pi}{3}$

d. $r = \sqrt{3}, \theta = \frac{4\pi}{3}$

First, express the base complex number in polar form. Let $w = \frac{\sqrt{3}}{2} + \frac{1}{2}i$. The modulus of w is $|w| = \sqrt{\left(\frac{\sqrt{3}}{2}\right)^2 + \left(\frac{1}{2}\right)^2} = \sqrt{\frac{3}{4} + \frac{1}{4}} = \sqrt{1} = 1$. The argument of w is $\arg(w) = \arctan\left(\frac{1/2}{\sqrt{3}/2}\right) = \arctan\left(\frac{1}{\sqrt{3}}\right)$. Since the real and imaginary parts are both positive, w is in the first quadrant, so $\arg(w) = \frac{\pi}{6}$. Thus.

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36 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

The graph of the function $y = f(x)$ is shown below. The function is of the form $f(x) = ax^2 + bx + c$. The vertex of the parabola is at $(2, -4)$. The parabola passes through the points $(0, 0)$ and $(4, 0)$. Determine the value of a.

a. **-1 ✓**

b. 1

c. -2

d. 2

The general form of a quadratic function is $f(x) = ax^2 + bx + c$. The vertex form of a quadratic function is $f(x) = a(x - h)^2 + k$, where (h, k) is the vertex. In this case, the vertex is $(2, -4)$, so $h = 2$ and $k = -4$. Thus, the function can be written as $f(x) = a(x - 2)^2 - 4$. We are given that the parabola passes through the point $(0, 0)$. Substitute these coordinates into the vertex form equation: $0 = a(0 - 2)^2 - 4$ $0 = a(-2)^2 - 4$ $0 = 4a - 4$ $4a = 4$ $a = 1$. Let's verify with the other given point.

37 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

A complex number z is represented in polar form as $z = r(\cos\theta + i\sin\theta)$. Given that $z^3 = 8i$, and z lies in the first quadrant such that its argument θ satisfies $0 < \theta < \frac{\pi}{2}$. Find the value of $r^2\cos\theta$.

a. 2

b. **4 ✓**

c. $2\sqrt{2}$

d. 1

We are given $z^3 = 8i$. In polar form, $8i = 8(\cos\frac{\pi}{2} + i\sin\frac{\pi}{2})$. Let $z = r(\cos\theta + i\sin\theta)$. Then $z^3 = r^3(\cos(3\theta) + i\sin(3\theta))$. Equating the magnitudes: $r^3 = 8$, so $r = 2$. Equating the arguments: $3\theta = \frac{\pi}{2} + 2k\pi$ for integer k. Since z is in the first quadrant, $0 < \theta < \frac{\pi}{2}$. For $k = 0$, $3\theta = \frac{\pi}{2}$, so $\theta = \frac{\pi}{6}$. This lies in the first quadrant. For $k = 1$.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

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38 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 1.04 Partial Fractions

A complex number z is represented in polar form as $z = r(\cos(\theta) + i\sin(\theta))$. If a complex number has a modulus $r = 5$ and an argument $\theta = \frac{\pi}{3}$, what is its rectangular form $a + bi$?

a. $\frac{5}{2} + i\frac{5\sqrt{3}}{2}$ ✓

b. $5 + i5$

c. $\frac{5\sqrt{3}}{2} + i\frac{5}{2}$

d. $5i$

The rectangular form of a complex number given in polar form $z = r(\cos(\theta) + i\sin(\theta))$ is $a + bi$, where $a = r\cos(\theta)$ and $b = r\sin(\theta)$. Given $r = 5$ and $\theta = \frac{\pi}{3}$: $a = 5\cos(\frac{\pi}{3}) = 5 \times \frac{1}{2} = \frac{5}{2}$ $b = 5\sin(\frac{\pi}{3}) = 5 \times \frac{\sqrt{3}}{2} = \frac{5\sqrt{3}}{2}$ Therefore, the rectangular form is $\frac{5}{2} + i\frac{5\sqrt{3}}{2}$.

39 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 1.05 Systems of Linear Equations (AA HL)

Find the oblique asymptote of $f(x) = \frac{4x^2 + 3x - 14}{(x + 2)}$, in the form $y = mx + c$.

a. $y = 4x - 5$ ✓

b. $y = 6x - 4$

c. $y = 5x - 4$

d. $y = 3x - 4$

Working through this functions step: Dividing : $\frac{4x^2 + 3x - 14}{(x + 2)} = 4x - 5 + \frac{-4}{(x + 2)}$, so the asymptote is $y = 4x - 5$.

40 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 2.01 Modulus Functions & the Factor/Remainder Theorem

Area enclosed by $r = 2\cos\theta$ (polar):

a. π ✓

b. 2π

c. 4π

d. $\pi/2$

Applying AA HL — Additional Topics principles, A=½∫■^π 4cos²θ dθ=π. Therefore, the verified correct option is π.

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41 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 3.01 3D Vectors & the Cross Product

Express $f(x) = \frac{x^2 - 3x + 5}{(x - 3)}$ in the form $mx + c + \frac{r}{(x - 3)}$, and state the value of r.

a. 8

b. 7

c. 2

d. **5** ✓

Polynomial division gives quotient x with remainder $r = 5$.

42 [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 3.02 Lines and Planes in Three Dimensions

The graph of the function $f(x) = ax^2 - 6x + c$ is shown below. The graph has a vertex at the point (3, -9) and passes through the point (0, 0). Determine the values of a and c.

a. $a = 1, c = 0$

b. $a = 2, c = 0$ ✓

c. $a = 1, c = 3$

d. $a = 2, c = 3$

The function is given by $f(x) = ax^2 - 6x + c$. The vertex of a parabola in the form $f(x) = Ax^2 + Bx + C$ is located at $x = -\frac{B}{2A}$. In this case, $A = a$ and $B = -6$. Therefore, the x-coordinate of the vertex is $x = -\frac{-6}{2a} = \frac{6}{2a} = \frac{3}{a}$. We are given that the vertex is at (3, -9), so the x-coordinate of the vertex is 3. Equating the two expressions for the x-coordinate of the vertex: $\frac{3}{a} = 3$ Multiplying both sides by a gives $3 = 3a$, so $a = 1$. Now, we use the y-c.

43 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 3.03 Compound Angle Identities & Reciprocal Trigonometric Functions (AA HL)

Consider the complex number $z = r(\cos\theta + i\sin\theta)$. If $z^3 = -8$ and $r > 0$, and the principal argument of z is $\frac{\pi}{3}$, find the value of r.

a. **2** ✓

b. 4

c. 8

d. 16

We are given the complex number $z = r(\cos\theta + i\sin\theta)$, where $r > 0$. We are also given that $z^3 = -8$ and the principal argument of z is $\theta = \frac{\pi}{3}$. Using De Moivre's Theorem, we have $z^3 = r^3(\cos(3\theta) + i\sin(3\theta))$. We are given $z^3 = -8$, which can be written in polar form as $8(\cos(\pi + 2k\pi) + i\sin(\pi + 2k\pi))$ for any integer k. Equating the magnitudes, we get $r^3 = 8$. Since $r > 0$, we have $r = \sqrt[3]{8} = 2$. Equating the arguments, we get.

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44 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 4.01 Unbiased Estimators & Confidence Intervals

The diagram shows the graph of the function $y = f(x)$. The function is defined by $f(x) = x^2 + bx + c$. The graph passes through the point $(1, 0)$ and has a vertex at $(2, -1)$. Find the value of b.

a. **-4 ✓**

b. 4

c. -2

d. 2

The vertex of a parabola given by the equation $y = ax^2 + bx + c$ is at $x = -\frac{b}{2a}$. In this case, $a = 1$. The vertex is given as $(2, -1)$, so the x-coordinate of the vertex is 2. Therefore, $2 = -\frac{b}{2(1)}$, which implies $2 = -\frac{b}{2}$. Multiplying both sides by 2 gives $4 = -b$, so $b = -4$. Alternatively, using the vertex form of the quadratic equation, $y = a(x - h)^2 + k$, where (h, k) is the vertex. Here, $a = 1$, $h = 2$, and $k = -1$. So, $y = 1(x - 2)^2 - 1$. Expanding this, we get.

45 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 4.02 The Poisson Distribution

The diagram shows the graph of the quadratic function $f(x) = ax^2 + bx - 2$. The graph passes through the points $(1, -3)$ and $(2, 2)$. Determine the value of a.

a. 1

b. **2 ✓**

c. 3

d. 4

The function is given by $f(x) = ax^2 + bx - 2$. Since the graph passes through $(1, -3)$, we have $f(1) = -3$: $a(1)^2 + b(1) - 2 = -3$ $a + b - 2 = -3$ $a + b = -1$ (Equation 1) Since the graph passes through $(2, 2)$, we have $f(2) = 2$: $a(2)^2 + b(2) - 2 = 2$ $4a + 2b - 2 = 2$ $4a + 2b = 4$ $2a + b = 2$ (Equation 2) Now we solve the system of linear equations for a and b. Subtract Equation 1 from Equation 2: $(2a + b) - (a + b) = 2 - (-1)$ $a = 3$ Substitute $a = 3$ into Equation 1: $3 + b = -1$ $b = -4$ T.

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